

Eigenvalues & Eigenvectors

The Characteristic Equation

- We know that A is invertible if and only if $\det A \neq 0$.
- This shows how to determine when a matrix of the form $A - \lambda I$ is not invertible.

Definition

The scalar equation $\det(A - \lambda I) = 0$ is called the **characteristic equation** of A , and the argument in Example 1 justifies the following fact.

A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

The Characteristic Equation

Theorem. *Let A be an $n \times n$ matrix, and let $f(\lambda) = \det(A - \lambda I_n)$ be its characteristic polynomial. Then $f(\lambda)$ is a polynomial of degree n . Moreover, $f(\lambda)$ has the form*

$$f(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} \operatorname{Tr}(A) \lambda^{n-1} + \cdots + \det(A).$$

In other words, the coefficient of λ^{n-1} is $\pm \operatorname{Tr}(A)$, and the constant term is $\det(A)$ (the other coefficients are just numbers without names).

Recipe: The characteristic polynomial of a 2×2 matrix. When $n = 2$, the previous theorem tells us all of the coefficients of the characteristic polynomial:

$$f(\lambda) = \lambda^2 - \operatorname{Tr}(A)\lambda + \det(A).$$

This is generally the fastest way to compute the characteristic polynomial of a 2×2 matrix.

The Characteristic Equation

EXAMPLE 7 Find the characteristic polynomial of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

SOLUTION

We have

$$f(\lambda) = \lambda^2 - \text{Tr}(A)\lambda + \det(A) = \lambda^2 - (5 + 1)\lambda + (5 \cdot 1 - 2 \cdot 2) = \lambda^2 - 6\lambda + 1$$

Find the characteristic equation of

$$\begin{pmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{pmatrix}$$

The characteristic equation is $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$

Where,

S_1 = sum of the main diagonal elements

$$= 2+1-4 = -1$$

S_2 = sum of minor of main diagonal elements

$$= \begin{vmatrix} 1 & 3 \\ 2 & -4 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ -5 & -4 \end{vmatrix} + \begin{vmatrix} 2 & -3 \\ 3 & 1 \end{vmatrix}$$

$$= (-4-6)+(-8+5)+(2+9) = -10+(-3)+11 = -2$$

S_3 = Det (A) = |A|

$$= \begin{vmatrix} 2 & -3 & 1 \\ 3 & 1 & 3 \\ -5 & 2 & -4 \end{vmatrix}$$

$$= 2(-4-6)-(-3)(-12+15)+1(6+5)$$

$$= 2(-10) + 3(3) + 1(11) = -20+9+11 = 0$$

The characteristic equation is $\lambda^3 + \lambda^2 - 2\lambda = 0$

Eigenvalues & Eigenvectors

Matrix:

Example 1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

The eigenvalues are obtained by solving the characteristic equation:

$$\det(A - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} 1 - \lambda & 0 \\ 0 & 1 - \lambda \end{bmatrix} \right) = (1 - \lambda)(1 - \lambda) = 0$$

$$\lambda_1 = 1, \lambda_2 = 1$$

For $\lambda = 1$, solve $(A - \lambda I)\mathbf{v} = 0$:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Any vector $\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$ is an eigenvector. Examples include:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Example 2

Matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = \det \left(\begin{bmatrix} 2 - \lambda & 0 \\ 0 & 3 - \lambda \end{bmatrix} \right) = (2 - \lambda)(3 - \lambda) = 0$$

$$\lambda_1 = 2, \lambda_2 = 3$$

For $\lambda = 2$:

$$(A - 2I)\mathbf{v} = 0 \implies \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

For $\lambda = 3$:

$$(A - 3I)\mathbf{v} = 0 \implies \begin{bmatrix} -1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

Eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Example 3

Matrix:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

$$\begin{aligned} \det(A - \lambda I) &= \det \left(\begin{bmatrix} 4 - \lambda & 1 \\ 2 & 3 - \lambda \end{bmatrix} \right) \\ &= (4 - \lambda)(3 - \lambda) - (2)(1) = \lambda^2 - 7\lambda + 10 = 0 \\ \lambda_1 &= 5, \lambda_2 = 2 \end{aligned}$$

For $\lambda = 5$:

$$(A - 5I)\mathbf{v} = 0 \implies \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

From the first row:

$$-x + y = 0 \implies y = x$$

Eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

For $\lambda = 2$:

$$(A - 2I)\mathbf{v} = 0 \implies \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0$$

From the first row:

$$-2x - y = 0 \implies y = -2x$$

Eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Example 3

Find all eigenvalues and corresponding eigenvectors for the matrix A if

$$\begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\begin{aligned} & \det \left(\begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \\ & \begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ & \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ & = \begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ & = \begin{pmatrix} 2-\lambda & -3 & 0 \\ 2 & -5-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{pmatrix} \\ & = \det \begin{pmatrix} 2-\lambda & -3 & 0 \\ 2 & -5-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{pmatrix} \\ & = (2-\lambda) \det \begin{pmatrix} -5-\lambda & 0 \\ 0 & 3-\lambda \end{pmatrix} - (-3) \det \begin{pmatrix} 2 & 0 \\ 0 & 3-\lambda \end{pmatrix} + 0 \cdot \det \begin{pmatrix} 2 & -5-\lambda \\ 0 & 0 \end{pmatrix} \\ & = (2-\lambda) (\lambda^2 + 2\lambda - 15) - (-3) \cdot 2(-\lambda + 3) + 0 \cdot 0 \\ & = -\lambda^3 + 13\lambda - 12 \\ & -\lambda^3 + 13\lambda - 12 = 0 \\ & -(\lambda - 1)(\lambda - 3)(\lambda + 4) = 0 \\ & \text{The eigenvalues are :} \\ & \lambda = 1, \lambda = 3, \lambda = -4 \end{aligned}$$

Example 4

Eigenvectors for $\lambda = 1$

$$\begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix} - 1 \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -3 & 0 \\ 2 & -6 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

$$(A - 1I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & -3 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} x - 3y = 0 \\ z = 0 \end{cases}$$

Isolate

$$\begin{cases} z = 0 \\ x = 3y \end{cases}$$

Plug into $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\eta = \begin{pmatrix} 3y \\ y \\ 0 \end{pmatrix} \quad y \neq 0$$

Let $y = 1$

$$\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$$

Similarly

Eigenvectors for $\lambda = 3$: $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

Eigenvectors for $\lambda = -4$: $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$

The eigenvectors for $\begin{pmatrix} 2 & -3 & 0 \\ 2 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$$= \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

Example 5

Find the eigenvalues and eigenvectors of

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix}$$

$$\begin{aligned} & \det \left(\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \\ & \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ & \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ & = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix} - \begin{pmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{pmatrix} \\ & = \begin{pmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & 9-\lambda \end{pmatrix} \\ & = \det \begin{pmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & 4 \\ 0 & 4 & 9-\lambda \end{pmatrix} \\ & = (2-\lambda) \det \begin{pmatrix} 3-\lambda & 4 \\ 4 & 9-\lambda \end{pmatrix} - 0 \cdot \det \begin{pmatrix} 0 & 4 \\ 0 & 9-\lambda \end{pmatrix} + 0 \cdot \det \begin{pmatrix} 0 & 3-\lambda \\ 0 & 4 \end{pmatrix} \\ & = (2-\lambda) (\lambda^2 - 12\lambda + 11) - 0 \cdot 0 + 0 \cdot 0 \\ & = -\lambda^3 + 14\lambda^2 - 35\lambda + 22 \\ & -\lambda^3 + 14\lambda^2 - 35\lambda + 22 = 0 \\ & -(\lambda - 1)(\lambda - 2)(\lambda - 11) = 0 \end{aligned}$$

The eigenvalues are :

$$\lambda = 1, \lambda = 2, \lambda = 11$$

Eigenvectors for $\lambda = 1$

$$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix} - 1 \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 4 \\ 0 & 4 & 8 \end{pmatrix}$$

$$(A - 1I) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{cases} x = 0 \\ y + 2z = 0 \end{cases}$$

Isolate

$$\begin{cases} x = 0 \\ y = -2z \end{cases}$$

Plug into $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\eta = \begin{pmatrix} 0 \\ -2z \\ z \end{pmatrix} \quad z \neq 0$$

Let $z = 1$

$$\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

Similarly

Eigenvectors for $\lambda = 2$: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

Eigenvectors for $\lambda = 11$: $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

The eigenvectors for $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{pmatrix}$

$= \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

Can the eigenvalue be zero?

Yes, the eigenvalue can be zero.

Can a singular matrix have eigenvalues?

Every singular matrix has a 0 eigenvalue.

How to find the eigenvalues of a square matrix A?

Use the equation $\det(A - \lambda I) = 0$ and solve for λ . Determine all the possible values of λ , which are the required eigenvalues of matrix A.

Example 6

Matrix:

$$A = \begin{bmatrix} 6 & 2 & 0 \\ 2 & 3 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

To find the eigenvalues, solve the characteristic equation $\det(A - \lambda I) = 0$:

$$\det(A - \lambda I) = \begin{vmatrix} 6 - \lambda & 2 & 0 \\ 2 & 3 - \lambda & 1 \\ 0 & 1 & 2 - \lambda \end{vmatrix}$$

Expanding the determinant:

$$\begin{aligned} \det(A - \lambda I) &= (6 - \lambda) \begin{vmatrix} 3 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} - 2 \begin{vmatrix} 2 & 1 \\ 0 & 2 - \lambda \end{vmatrix} \\ &= (6 - \lambda)((3 - \lambda)(2 - \lambda) - 1) - 2(2(2 - \lambda)) \end{aligned}$$

Simplify:

$$\begin{aligned} \det(A - \lambda I) &= (6 - \lambda)((3 - \lambda)(2 - \lambda) - 1) - 4(2 - \lambda) \\ &= (6 - \lambda)(6 - 5\lambda + \lambda^2 - 1) - 8 + 4\lambda \\ &= \lambda^3 - 11\lambda^2 + 30\lambda - 22 \end{aligned}$$

$$\lambda_1 = 1, \lambda_2 = 4, \lambda_3 = 6$$

For $\lambda_1 = 1$:

Solve $(A - \lambda I)\mathbf{v} = 0$:

$$\begin{bmatrix} 5 & 2 & 0 \\ 2 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From the first row:

$$5x + 2y = 0 \implies y = -\frac{5}{2}x$$

From the second row:

$$2x + 2y + z = 0 \implies 2x - 5x/2 + z = 0 \implies z = x$$

Eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -\frac{5}{2} \\ 1 \end{bmatrix} \text{ (scaled to simplify if needed)}$$

For $\lambda_2 = 4$:

Solve $(A - 4I)\mathbf{v} = 0$:

$$\begin{bmatrix} 2 & 2 & 0 \\ 2 & -1 & 1 \\ 0 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From the first row:

$$2x + 2y = 0 \implies y = -x$$

From the second row:

$$2x - y + z = 0 \implies 2x + x + z = 0 \implies z = -3x$$

Eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}$$

For $\lambda_3 = 6$:

Solve $(A - 6I)\mathbf{v} = 0$:

$$\begin{bmatrix} 0 & 2 & 0 \\ 2 & -3 & 1 \\ 0 & 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From the first row:

$$2y = 0 \implies y = 0$$

From the second row:

$$2x + z = 0 \implies z = -2x$$

Eigenvector:

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

Example 7

Matrix:

$$A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Solve $\det(A - \lambda I) = 0$:

$$A - \lambda I = \begin{bmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & -1 & 0 \\ 1 & -\lambda & 0 \\ 0 & 0 & 1 - \lambda \end{vmatrix}$$

Expanding along the third row:

$$\begin{aligned} \det(A - \lambda I) &= (1 - \lambda) \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} \\ &= (1 - \lambda)(\lambda^2 + 1) \end{aligned}$$

Eigenvalues:

$$\lambda = 1, \lambda = i, \lambda = -i$$

For $\lambda = 1$:

Solve $(A - I)\mathbf{v} = 0$:

$$A - I = \begin{bmatrix} -1 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

The system becomes:

$$-x - y = 0 \quad \Rightarrow \quad y = -x.$$

Eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

For $\lambda = i$:

Solve $(A - iI)\mathbf{v} = 0$:

$$A - iI = \begin{bmatrix} -i & -1 & 0 \\ 1 & -i & 0 \\ 0 & 0 & 1 - i \end{bmatrix}$$

Focus on the 2×2 top-left block:

$$(-i)x - y = 0 \quad \Rightarrow \quad y = ix.$$

Eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}$$

For $\lambda = -i$:

Solve $(A + iI)\mathbf{v} = 0$:

$$A + iI = \begin{bmatrix} i & -1 & 0 \\ 1 & i & 0 \\ 0 & 0 & 1 + i \end{bmatrix}$$

Focus on the 2×2 top-left block:

$$(ix) - y = 0 \quad \Rightarrow \quad y = -ix.$$

Eigenvector:

$$\mathbf{v}_3 = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}$$

Principal Stress With Eigenvalues & Eigenvectors

In mechanics, **principal stresses** are determined from the stress tensor. This involves finding eigenvalues (principal stresses) and eigenvectors (principal directions) of the **stress matrix**.

Consider a **stress tensor** in a 3D state of stress:

$$\sigma = \begin{bmatrix} 100 & 30 & 0 \\ 30 & 50 & 0 \\ 0 & 0 & 20 \end{bmatrix} \text{ (in MPa)}$$

Here:

- Diagonal elements ($\sigma_{xx}, \sigma_{yy}, \sigma_{zz}$) are normal stresses.
- Off-diagonal elements ($\tau_{xy}, \tau_{xz}, \tau_{yz}$) are shear stresses.

The goal is to find the **principal stresses** (eigenvalues) and their associated **principal directions** (eigenvectors).

Find Principal Stresses (Eigenvalues)

$$\det(\sigma - \lambda I) = 0$$

$$\det \begin{bmatrix} 100 - \lambda & 30 & 0 \\ 30 & 50 - \lambda & 0 \\ 0 & 0 & 20 - \lambda \end{bmatrix} = 0$$

The determinant expands as:

$$\begin{aligned} & (100 - \lambda) \begin{vmatrix} 50 - \lambda & 0 \\ 0 & 20 - \lambda \end{vmatrix} - 30 \begin{vmatrix} 30 & 0 \\ 0 & 20 - \lambda \end{vmatrix} \\ &= (100 - \lambda)((50 - \lambda)(20 - \lambda)) - (30)(30)(20 - \lambda) \end{aligned}$$

Simplify:

$$= (100 - \lambda)((50 - \lambda)(20 - \lambda)) - 900(20 - \lambda)$$

Expand $(50 - \lambda)(20 - \lambda)$:

$$(100 - \lambda)(1000 - 70\lambda + \lambda^2) - 900(20 - \lambda)$$

Combine terms:

$$\lambda^3 - 170\lambda^2 + 7850\lambda - 130000 = 0$$

Solve for λ (eigenvalues), which are:

$$\lambda_1 = 120 \text{ MPa}, \quad \lambda_2 = 40 \text{ MPa}, \quad \lambda_3 = 20 \text{ MPa}$$

These are the **principal stresses**.

Find Principal Directions (Eigenvectors)

For $\lambda_1 = 120$:

$$\sigma - \lambda_1 I = \begin{bmatrix} -20 & 30 & 0 \\ 30 & -70 & 0 \\ 0 & 0 & -100 \end{bmatrix}$$

Solve:

$$\begin{aligned} -20x + 30y &= 0 &\Rightarrow & y = \frac{2}{3}x, \\ z &= 0. \end{aligned}$$

Eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} \quad (\text{normalized: } \mathbf{v}_1 = \frac{1}{\sqrt{13}} \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}).$$

For $\lambda_2 = 40$:

$$\sigma - \lambda_2 I = \begin{bmatrix} 60 & 30 & 0 \\ 30 & 10 & 0 \\ 0 & 0 & -20 \end{bmatrix}$$

Solve:

$$\begin{aligned} 60x + 30y &= 0 &\Rightarrow & y = -2x, \\ z &= 0. \end{aligned}$$

Eigenvector:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} \quad (\text{normalized: } \mathbf{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}).$$

For $\lambda_3 = 20$:

$$\sigma - \lambda_3 I = \begin{bmatrix} 80 & 30 & 0 \\ 30 & 30 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Solve:

$$80x + 30y = 0 \quad \Rightarrow \quad y = -\frac{8}{3}x, \\ z = 1.$$

Eigenvector:

$$\mathbf{v}_3 = \begin{bmatrix} 3 \\ -8 \\ 0 \end{bmatrix} \quad (\text{normalized: } \mathbf{v}_3 = \frac{1}{\sqrt{73}} \begin{bmatrix} 3 \\ -8 \\ 0 \end{bmatrix}).$$

- Principal Stresses:

$$\sigma_1 = 120 \text{ MPa}, \sigma_2 = 40 \text{ MPa}, \sigma_3 = 20 \text{ MPa}.$$

- Principal Directions:

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ -8 \\ 0 \end{bmatrix}.$$

The principal stresses correspond to the eigenvalues, while the principal directions (eigenvectors) describe the orientation of the principal stress axes.